Probabilities

1. A fair die with faces numbered from 1 to 6 is rolled. Calculate the probability that the die lands with an even number uppermost?

Number of faces on the dice with even numbers/ number of faces on the dice = 3/6 =1/2

2. Coronary heart disease (CHD) is currently the most common cause of death in the UK. In 2002, of 288 332 male deaths, 64 473 were from CHD, and of 318 463 female deaths, 53 003 were from CHD.

- Estimate the following probabilities.
    - The probability that a randomly chosen UK man who dies in a future year will die of CHD.
    - The probability that a randomly chosen UK woman who dies in a future year will die of CHD.

i) number of male deaths from CHD/ number of male deaths = 64473/288332 = 0.2236

ii) Number of female deaths from CHD/ number of female deaths = 53003/318463 = 0.1664

How accurate are your estimates likely to be?

I predict my estimates to be quite good as they are based on a large amount of data from 2002, this is assuming that the probability from CHD has not changed since 2002.